

Expense Tracker

Business Requirements · v2.0

1. What to build

A REST API for company expense claims. Employees record expenses and group them into claims. Approvers approve or reject claims. Finance maintains categories and marks approved claims as reimbursed.

This document defines the entities, operations and rules. It does not define endpoints, paths, HTTP methods or status codes — those are design decisions.

Three entities only: **Category**, **Expense**, **Claim**. Reporting currency is EUR. All monetary values are exact to 2 decimal places.

2. Entities

Category

FIELD	TYPE	REQ	RULES
id	string	gen	Server-generated. Immutable.
name	string	yes	1–50 characters. Unique across categories.
receiptThreshold	decimal	yes	EUR. A receipt is required when an expense in this category exceeds it. 0 means always required.
active	boolean	no	Defaults to true. Only active categories may be assigned to a new expense.

Expense

FIELD	TYPE	REQ	RULES
id	string	gen	Server-generated. Immutable.
ownerId	string	gen	The authenticated employee who created it. Immutable.
amount	decimal	yes	Must be > 0. The amount as paid, in currency.
currency	string	yes	ISO 4217 three-letter code, uppercase. Example: USD.
amountEur	decimal	yes	Must be > 0. The same amount in EUR. Equals amount when currency is EUR.
dateIncurred	date	yes	ISO 8601 date (YYYY-MM-DD). Must not be in the future.
categoryId	string	yes	Must reference an existing, active Category.
description	string	yes	1–200 characters.
paymentMethod	enum	yes	One of: personal, company_card.
clientReference	string	no	Free text, max 50 characters.
receipts	array	no	0 to 10 receipt references. JPEG, PNG or PDF, max 10 MB each.
claimId	string	no	Null until the expense is added to a Claim.
createdAt	timestamp	gen	ISO 8601. Server-generated. Immutable.

Claim

FIELD	TYPE	REQ	RULES
id	string	gen	Server-generated. Immutable.
ownerId	string	gen	The authenticated employee who created it. Immutable.
title	string	yes	1–100 characters.
expenseIds	array	yes	References to Expenses owned by the same employee.
totalEur	decimal	gen	Computed: the sum of amountEur of the expenses it contains.

FIELD	TYPE	REQ	RULES
status	enum	gen	One of: draft, submitted, approved, rejected, reimbursed. Starts at draft.
submittedAt	timestamp	no	Set when status becomes submitted. Null before that.
decidedBy	string	no	The approver who approved or rejected it.
decidedAt	timestamp	no	Set when status becomes approved or rejected.
rejectionReason	string	no	1–500 characters. Required when rejecting. Preserved after the claim returns to draft.

3. Roles and required operations

The caller's role is carried in the access token. Anything not listed here must be refused for that role.

ROLE	MUST BE ABLE TO
Employee	Create, read, update and delete their own expenses. List their own expenses, filtered by date range, category, and whether the expense is already in a claim. Attach and remove receipts on their own expenses. Read the category list. Create a claim, add and remove their own expenses from it while it is in draft, and submit it. List their own claims and read one.
Approver	List claims that are submitted and not their own. Read one such claim including its expenses and receipts. Approve a claim. Reject a claim with a reason.
Finance	Create, rename, retire and list categories, and change a receipt threshold. Mark an approved claim as reimbursed. List all expenses across all employees for a date range. Read any claim.

4. Rules

Every rule below must be enforced. A request that breaks one must be refused with a response identifying which rule failed.

ID	RULE
R1	amount and amountEur must be greater than 0. Negative amounts and refunds are not supported.
R2	dateIncurred must not be later than today.
R3	categoryId must reference a Category whose active is true at the time the expense is created or its category is changed.
R4	An expense whose dateIncurred is more than 90 days before today cannot be added to a claim. It can still be created, read and updated.
R5	An expense whose amountEur exceeds its category's receiptThreshold must have at least one receipt before its claim can be submitted.
R6	A claim must contain at least one expense to be submitted.
R7	An expense can belong to at most one claim. Adding it to a second claim must be refused.
R8	All expenses in a claim must have the same ownerId as the claim.
R9	Claim status may change only as follows, and by no other path: draft → submitted; submitted → approved; submitted → rejected; approved → reimbursed; rejected → draft.
R10	Rejecting a claim requires rejectionReason. Approving must not accept one.
R11	While a claim's status is submitted, approved or reimbursed, neither the claim nor any expense in it may be modified or deleted, and expenses may not be added or removed.
R12	An approver must not approve or reject a claim whose ownerId is their own.
R13	A claim with status reimbursed is final. No further change of any kind is permitted.
R14	Retiring a category must not change or invalidate any existing expense assigned to it.
R15	An employee must not read, modify or delete an expense or claim owned by another employee.
R16	totalEur is always derived from the claim's expenses and must never be accepted from the caller.

5. Technical requirements

ID	RULE
T1	Every request must be authenticated. No operation may be reachable anonymously.
T2	Authentication uses a JWT bearer token. The token carries the caller's identity and one role: employee, approver or finance.
T3	An authenticated caller attempting an operation outside their role must be refused explicitly, not given an empty result.
T4	Every list operation must be paginated, with a caller-supplied page size capped at a documented maximum.
T5	Every error response must identify the rule that failed in a form a client can display to a user.
T6	Every operation must have at least one success example and one failure example documented.

6. Out of scope

Corporate card feeds and transaction matching · reading data from receipt images · mileage calculation · executing payments · budget enforcement · currency conversion (the caller supplies amountEur) · multiple approvers on one claim.